

● *Original Contribution***EFFECTS OF NONLINEAR PROPAGATION IN ULTRASOUND CONTRAST AGENT IMAGING**MENG-XING TANG,^{*} NAOHISA KAMIYAMA,[†] and ROBERT J. ECKERSLEY[‡]^{*}Department of Bioengineering, Faculty of Engineering, Imperial College London, UK; [†]Application & Research Group, Ultrasound Division, Toshiba Medical Systems Corp., Otawara, Japan; and [‡]Imaging Sciences Department, Faculty of Medicine, Imperial College London, UK

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Abstract—This paper investigates two types of nonlinear propagation and their effects on image intensity and contrast-to-tissue ratio (CTR) in contrast ultrasound images. Previous studies have shown that nonlinear propagation can occur when ultrasound travels through tissue and microbubble clouds, making tissue farther down the acoustic path appear brighter in pulse inversion (PI) images, thus reducing CTR. In this study, the effect of nonlinear propagation through tissue or microbubbles on PI image intensity and CTR are compared at low mechanical index. A combination of simulation and experiment with SonoVue microbubbles were performed using a microbubble dynamics model, a laboratory ultrasound system and a clinical prototype scanner. The results show that, close to the bubble resonance frequency, nonlinear propagation through a bubble cloud of a few centimeter thickness with a modest concentration (1:10000 dilution of SonoVue microbubbles) is much more significant than through tissue-mimicking material. Consequently, CTR in regions distal to the imaging probe is greatly reduced for nonlinear propagation through the bubble cloud, with as much as a 12-dB reduction compared with nonlinear propagation through tissue-mimicking material. Both types of nonlinear propagation cause only a small change in bubble PI signals at the bubble resonance frequency. When the driving frequency increases beyond bubble resonance, nonlinear propagation through bubbles is greatly reduced in absolute values. However because of a greater reduction in nonlinear scattering from bubbles at higher frequencies, the corresponding CTR is much lower than that at bubble resonance frequency. (E-mail: mengxing.tang@imperial.ac.uk) © 2010 World Federation for Ultrasound in Medicine & Biology.

Key Words: Contrast agents, Nonlinear propagation, Imaging artefacts, Perfusion quantification.

INTRODUCTION

Microbubbles have shown great potential as ultrasound (US) contrast agents in various clinical applications in cardiology and radiology, and recently these have been extended to targeted imaging and drug/gene delivery (Ferrara et al. 2007). Advanced techniques such as pulse inversion (PI) (Chapman and Lazenby 1997; Averkiou 2001), pulse inversion Doppler (Simpson et al. 1999), amplitude modulation (Brock-Fisher et al. 1996), dual-frequency modulation (Bouakaz et al. 2007; Burns 2007) and other complex pulse sequences (Borsboom et al. 2005; Chetty et al. 2006; Eckersley et al. 2007) have been developed to improve the sensitivity and specificity when imaging these microbubbles. These techniques are built on the basis that microbubbles can oscillate nonlinearly under US excitation and produce nonlinear echoes. Each

approach aims to extract these nonlinear echoes while suppressing linear echoes from tissue, to produce contrast-specific images. In these images, the ratio of the signals from microbubbles to that from surrounding tissue is a measure of the sensitivity of the technique and is termed the contrast-to-tissue ratio (CTR) (Bouakaz et al. 2002). Such images are often overlaid on standard B-mode images for anatomical reference. Quantification of tissue perfusion can be obtained based on the intensity of these contrast-specific images (Becher and Burns 2000; Bhatia and Senior 2008; Lavis et al. 2008; Wei et al. 1998). However, the accuracy and consistency of such quantification has been affected by a number of factors (Stride et al. 2009), attenuation being one which has been the subject of a few recent studies (Yano et al. 2004; Mule et al. 2008; Tang et al. 2008).

In addition to the nonlinear scattering described above, US is also known to propagate nonlinearly. Nonlinear propagation of US can occur in tissues and through microbubble clouds. Nonlinear propagation

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through tissue has been studied extensively and used successfully in diagnostic imaging. Reviews on this topic can be found in [Hamilton and Blackstock \(1998\)](#), [Duck \(2002\)](#) and [Humphrey \(2000\)](#). For nonlinear propagation through bubbly medium, a notable work is that of [Caflish et al. \(1985\)](#), who described a model consisting of a system of effective equations for wave propagation in a bubbly liquid. [Commander and Prosperetti \(1989\)](#) compared this model with experimental measurements and found the model to be accurate when most bubbles are not at resonance. [de Jong \(1993\)](#) has measured the nonlinearity parameter B/A for a microbubble solution and found it to be much higher than water. Despite these efforts, nonlinear propagation through bubbly medium when bubbles are resonant is still not fully understood, and discrepancies between existing models and experimental measurements remain ([Hibbs et al. 2007](#); [Tang and Eckersley 2006](#)). Clinical situations where significant nonlinear propagation may exist include contrast echocardiography, where US pulses can become significantly nonlinear after propagating through the microbubble-filled ventricles of the heart. In contrast-enhanced liver imaging, nonlinear propagation may become significant beyond a certain depth, especially because the blood-rich liver may contain a large amount of microbubbles.

During image acquisition, the nonlinear distortions to the propagating US pulses will be backscattered by bubbles and tissue as the distorted pulses continue to propagate. Current image formation algorithms cannot distinguish the nonlinear signals due to propagation from those arising solely from microbubble nonlinear scattering. Consequently, imaging artifacts appear and contrast is reduced. A previous study ([Tang and Eckersley 2006](#)) has shown that nonlinear propagation is able to create two types of imaging artifact: increasing nonlinear scattering from tissues (Error Type-1) and changing nonlinear scattering from bubbles (Error Type-2). Error Type-1 reduces CTR by increasing the denominator, whereas Error Type-2 can potentially influence the numerator both ways.

A few studies have reported the effects of nonlinear propagation on the behavior of microbubbles. [Ayme and Carstensen \(1989\)](#) and [Ayme-Bellegarda \(1990\)](#) have examined the response of microbubbles in terms of transient cavitation, collapse and rebound caused by nonlinearly distorted pulses. They concluded that higher-order harmonics introduced in the nonlinearly distorted pulse are relatively unimportant in the occurrence of these behaviors. [Kvikliene et al. \(2004\)](#) measured nonlinear propagation through water and input the results to a microbubble simulator to estimate the effect that incident pulse distortion as a result of nonlinear propagation would have on microbubble nonlinear scattering. They showed an increase (2~3 dB or 25~40%) of nonlinear scattering

from bubbles because of nonlinear propagation through water. Both studies used acoustic pressures much higher than that typically used clinically.

Despite previous studies on the topic of US nonlinear propagation, many questions remain unanswered in the context of contrast agent imaging. Notably, it is not clear to what extent the nonlinear propagation of US affects the image intensity and the CTR at the low mechanical index (MI) typically used for contrast US imaging. Furthermore, it is also not clear how the effects of nonlinear propagation will vary with the type of medium within which nonlinear propagation occurs and the imaging system parameters such as frequency and MI. This paper extends our previous work by investigating the effect of nonlinear propagation as a result of tissues and bubbles on the image intensity and CTR in US contrast agent images.

METHODS

Transmission measurements

Transmission setup. A laboratory US rig was used ([Fig. 1](#)) to measure US pulses propagated through two different media. The measured signals were used as input for a numerical simulation of microbubble scattering. A broadband single-element transducer (Videoscans V380, radius 25 mm, point focus at 75 mm, Panametrics-NDT, Waltham, MA, USA) was used to transmit a series of US pulses at 4 frequencies (1, 2, 4, 5.7 MHz). It should be noted that the Panametrics transducer may have nonlinear distortion at the lowest (1 MHz) and highest (5.7 MHz) frequencies and this should be taken into account when interpreting the results.

Each pulse consisted of 4 cycles with a Gaussian envelop ([Fig. 2](#)). The transmitted signals were measured using a broadband needle hydrophone (HPM 05/3, radius 0.25 mm, Precision Acoustics Ltd, Dorset UK), which was positioned at the focus of the transmission transducer. This position was 77 ± 1 mm from the transducer surface, determined by a measurement at 5.7 MHz and fixed for all measurements. For each frequency, three pulses were transmitted with measured peak-positive pressure at focus ranging from 50~100 kPa at 1 MHz and 120~238 kPa at 5.7 MHz. This range of pressure roughly corresponds to an MI range of 0.05 – 0.1 at each frequency, equivalent

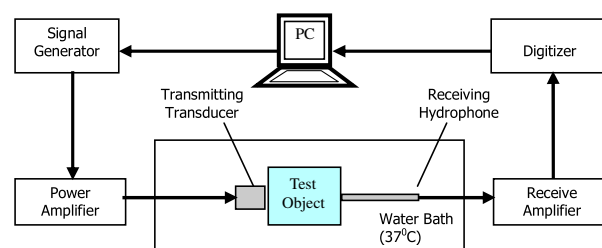


Fig. 1. Experimental setup for transmission measurements.

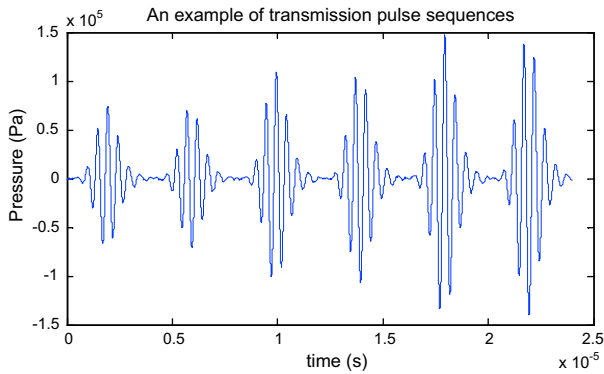


Fig. 2. An example of a transmitted pulse sequence at 2 MHz: PI pulse pairs with three power ramps. Note that the time intervals between neighboring pulses have been reduced compared with that used in the experiment to display the pulses clearly.

to that used clinically for minimum microbubble destruction. Test objects consisting of either a container holding microbubbles or a block of tissue-mimicking material (TMM) were inserted between the transducer and the hydrophone. To avoid any nonlinearity buildup outside the test object, they were placed close to the hydrophone (~ 1 mm). As the attenuation because of TMM is slightly different from that of microbubble clouds, the power of the driving pulses was adjusted for each of the test objects so that the measured pressure at focus was held constant at each frequency. Each transmission pulse was followed immediately by an inverted pulse to make up a PI pair. An example of one pulse sequence is shown in Fig. 2. The repetition rate of the pulses was 200 Hz.

Microbubbles and TMM. SonoVue (Bracco S.p.A., Milan, Italy) microbubbles were diluted to a modest concentration (100 μ L of SonoVue per liter [1:10000]) in saline at 37°C. The diluted microbubbles were contained in a beaker with acoustic windows on both sides and an acoustic path length of 6 cm. The windows were made from 1.5- μ m-thick polyethylene terephthalate (Mylar, Goodfellow Cambridge Ltd., Huntingdon, UK). The beaker was positioned between the transducer and the hydrophone within a water bath of 37°C. The microbubble suspension was gently stirred before every transmission measurement. TMM was made with agar, glycerin and graphite scatterers based on the formula described in the study by Teirlinck *et al.* (1998). The dimension of the TMM was $6.5 \times 4 \times 4$ cm.

Simulation of microbubble response

Simulation. The transmission measurements from the previous section were used as inputs to a microbubble dynamics simulator and the microbubble response to these propagated pulses was estimated. The inputs included the pulses after propagation through tissue or microbubbles

over a range of frequencies and MIs. The calculated microbubble scattering signals of pulse inversion pairs were summed to generate PI signals. To evaluate changes in PI signals as a result of nonlinear propagation, simulation results from undistorted linear pulses were generated as references. The linear pulses were also obtained from transmission measurements through water; however, in this case, low excitation powers were used. In addition, the measured signals were filtered to remove any harmonic components in the linear pulse measurements. To provide a fair comparison, the linear pulses were scaled to have the equivalent pulse power as the nonlinearly distorted pulses. The pulse power is defined here as the integral of the pressure squared for the pulse (*i.e.*, the digital samples of the pulse pressure signal are squared and then summed).

The simulation was based on the Rayleigh, Plesset, Noltingk, Neppiras, and Poritsky (RPNNP) model (de Jong *et al.* 1994) and the microbubble shell parameters follow those of Gorce (2000). A microbubble of 6 μ m in diameter was used in all simulations because this size is most common in the SonoVue population when measured in terms of gas volume (Gorce 2000). Details of other parameters used in the simulation can be found in Tang and Eckersley (2006).

Relative change in PI signal because of nonlinear propagation. Based on the simulation results, a quantity, termed here as *relative change* (RC), was obtained through dividing the power of the bubble PI signal generated from the nonlinearly distorted pulse pair by that from the linear pulse pair. The RC was then calculated for different frequencies.

Calculating CTR based on simulation. In addition to the PI signal calculated from bubble scattering, PI signals were also obtained through summing the nonlinearly distorted driving pulse pair. This sum is representative of tissue PI signals because the tissue linearly reflects the nonlinearly distorted pulses. Therefore, through dividing the PI signal from bubble scattering by that from the driving pulse pair, a quantity proportional to CTR was obtained.

In vitro phantom imaging with a commercial scanner

A phantom was constructed (Fig. 3) to experimentally evaluate the image CTR in the presence of nonlinear propagation. The phantom was housed in a beaker and all sides of the beaker were lined with acoustic absorption material (Aptflex F28, Precision Acoustics Ltd.). Diluted SonoVue suspension was introduced into the beaker. The dilution of the SonoVue microbubbles was 100 μ L/L of saline. The phantom was insonified from the top. A total of six regions-of-interest (ROI), shown as numbers in Fig. 3, were selected. This allowed the calculation of CTR in three areas of the image. These consisted of one area with relatively less nonlinear propagation (ROI 1, 2),

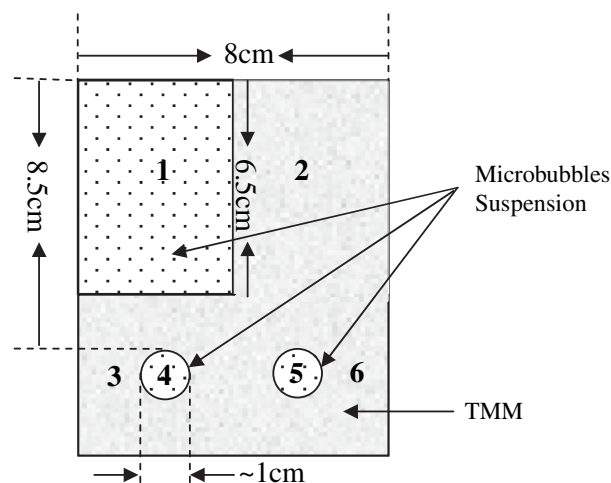


Fig. 3. *In vitro* imaging phantom consisting of TMM (grey areas) and SonoVue bubble suspension (dotted areas) with a concentration of 100 $\mu\text{L/L}$. The US probe was positioned at the top of the phantom.

a second with nonlinear propagation through microbubble clouds (ROI 3, 4) and a third with nonlinear propagation because of tissue (ROI 5, 6). For ROI 1 and 2, ideally regions closer to the probe would be chosen to provide CTR without any nonlinear propagation. However, this was not possible because the signal-to-noise ratio (SNR) in that area was relatively poor.

A Toshiba prototype scanner (AplioXG, Toshiba Medical Systems Corp, Otawara, Japan) with a convex transducer (PVT382BT) was used to scan the phantom in contrast harmonic imaging (CHI) mode. This uses a PI pulse pair sequence. Two types of transmission/reception conditions were used: (1) 1.75 MHz/3.5 MHz and (2) 2.5 MHz/5.0 MHz. 1.75 MHz is close to the resonance frequency of SonoVue microbubbles (Tang and Eckersley 2007). Although 2.5 MHz is not an ideal off-resonance frequency for SonoVue, it is the highest frequency that can be achieved with this probe for this imaging mode. Two levels of MI were used: $\text{MI} = 0.05$ and $\text{MI} = 0.08$. No higher MI was used because microbubble disruption was clearly visible at $\text{MI} = 0.1$. The transmission focus was always set to be at 6 cm. Radiofrequency (RF) data were acquired through customized software, based on which PI images were constructed and image CTR were calculated and compared. It should be noted that all the ROI analyses were based on linear data obtained by directly processing the acquired RF acoustic data without logarithmic compression.

RESULTS

Transmission measurements

Figure 4a shows the power of the PI signal as a function of driving acoustic pressure, after transmission through either bubble or the TMM at four different

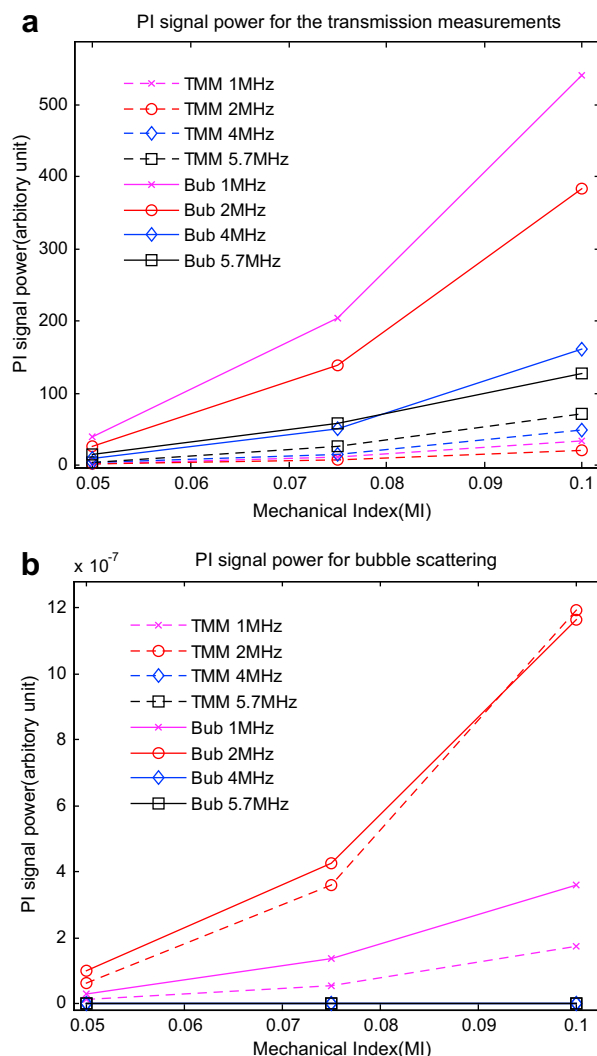


Fig. 4. (a) The power of PI signals for the transmission measurements through microbubbles or TMM as a function of MI at different frequencies; (b) the power of PI signals of the simulated scattering from bubbles as a function of MI at different frequencies.

frequencies. It can be seen that the PI power for nonlinear propagation through bubbles (solid lines) is significantly higher than the TMM (dotted lines) case. The PI power generally increases with frequency for nonlinear propagation through TMM. However, for nonlinear propagation through bubbles the PI signal power is much higher for 1 and 2 MHz.

Simulation results using nonlinearly propagated pulses

PI signal. The received PI signal powers from the simulation of bubble scattering are presented in Fig. 4b. The 2-MHz pulses show the highest PI signals (circles). At 1 MHz, some PI signals (crosses) are also detected. However, at 4 MHz or 5.7 MHz, the PI signals produced by scattering bubbles are much lower.

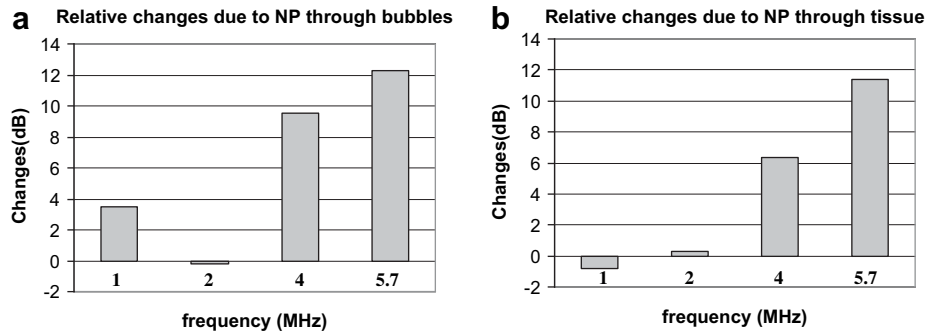


Fig. 5. Relative changes of calculated PI signals from bubbles because of nonlinear propagation through tissue (a) and microbubble clouds (b) at MI = 0.1. Negative dB values mean PI signals are reduced because of nonlinear propagation. It should be noted that the quantities presented in this figure are ratios and the denominators of the ratios are much smaller at 4 MHz and 5.7 MHz.

Relative changes. Fig. 5 shows the relative percentage changes in bubble PI signals because of the two types of nonlinear propagation. It can be seen that nonlinear propagation affects the microbubble nonlinear scattering most when away from the resonant frequency. At frequencies higher than the resonance frequency, the large positive values in Fig. 5 reveal that the nonlinear propagation makes bubbles produce relatively higher levels of nonlinear echoes and make them appear brighter than they should be in PI images. Closer to the resonant frequency at 2 MHz, the effect of nonlinear propagation is relatively small (<10% changes in this case). The small negative value in Fig. 5a for 2 MHz means that bubbles appear slightly darker at resonance frequency after nonlinear propagation through bubbles.

CTR as a function of frequency with the presence of two types of nonlinear propagation. Figure 6 shows the CTR of regions after nonlinear propagation through tissue (grey bars) or microbubble clouds (dark grey bars) as a function of frequency. The results show that the CTR

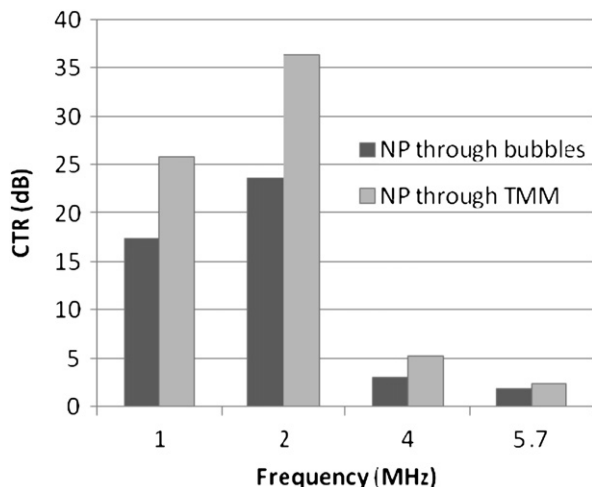


Fig. 6. Comparison of CTR for two types of nonlinear propagation at different frequencies.

after nonlinear propagation through microbubbles is consistently lower than that after nonlinear propagation through TMM. It should be noted that only a modest bubble concentration was used in this study. The difference in CTR between the two types of nonlinear propagation is most significant (12 dB) at 2 MHz. It can also be seen that CTR decreases significantly as driving frequency moves away from bubble resonance.

Results of in vitro phantom scan

Figure 7 shows the results of *in vitro* scan of the phantom described in Fig. 3. The CTRs at three areas in the images were calculated and shown in Table 1. First, it can be seen that at 1.75 MHz, which is the resonance frequency of SonoVue, the CTR for the area under the large bubble cloud (ROI4/ROI3) is significantly lower than that under the TMM (ROI5/ROI6). At 2.5 MHz, the difference in CTR between the two areas is much smaller. These correspond well with the simulation results presented in Fig. 6. Second, there is in general a reduction in CTR as the frequency is increased from 1.75 – 2.5 MHz, with the exception of ROI4/ROI3 under the bubble cloud at an MI of 0.08. The reason for this is discussed later. Furthermore, there seems to be a general reduction of CTR as MI increases from 0.05 – 0.08. However no further measurements at higher MIs were conducted because bubble disruption became apparent for MIs higher than 0.08. It should be noted that change to CTR as a function of MI depends on the relative significance between two quantities: PI signal from bubble scattering (Fig. 4b) and PI signal from nonlinear distortion during nonlinear propagation (Fig. 4a). These quantities are complex functions of the properties of bubbles, the medium and the parameters of the US system.

DISCUSSION

The aim of this study was to compare and quantify the effects of the two types of nonlinear propagation on

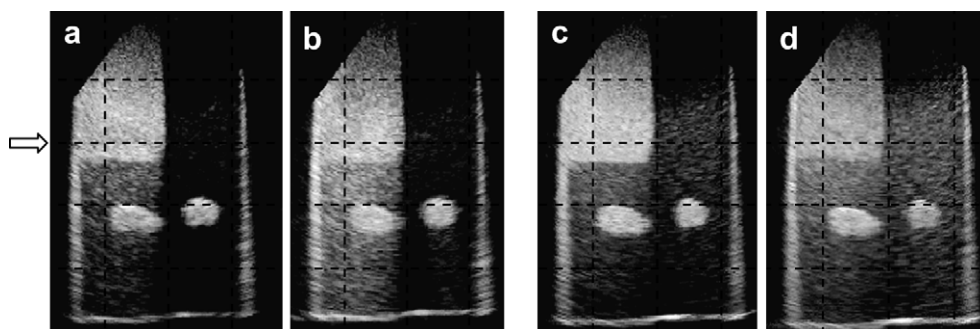


Fig. 7. PI images of the *in vitro* phantom using a Toshiba prototype scanner: (a) Frequency = 1.75 MHz, MI = 0.05; (b) frequency = 1.75 MHz, MI = 0.08; (c) frequency = 2.5 MHz, MI = 0.05; (d) frequency = 2.5 MHz, MI = 0.08.

the image intensity and CTR in PI images and the following findings have been observed.

Firstly, the results show that at low MIs and close to bubble resonance frequency, nonlinear propagation through a bubble cloud of even a modest concentration (1:10000 dilution in the case of SonoVue microbubbles) is much more significant than through TMM (Fig. 4a). Consequently, CTR in distal regions is greatly reduced for nonlinear propagation through the bubble cloud, with a 12-dB further reduction than that through TMM in this study (Fig. 6). It should be noted that the concentration of SonoVue used in this study was chosen to be representative of real clinical scans. For example, in contrast liver imaging, a typical clinical dose of 2.4 μL SonoVue is injected into a human body, which on average contains 6 L of blood. In this case, the dilution of SonoVue bubbles in blood is 4:10000. Because blood consists of nearly 30% of the liver volume, the concentration of bubbles used in this study would be similar to that within perfused liver tissue if removal of microbubbles, *e.g.*, through pulmonary circulation, is not significant. For a bolus injection, it is likely that the bubble concentration in liver may well exceed that used in this study, even if some bubbles are removed by organs such as the lungs.

Secondly, when the driving frequency increases beyond bubble resonance, nonlinear propagation through bubbles is reduced considerably (Fig. 4a), but its relative effects are greater (Fig. 5) because of the greater reduction of PI signals from bubbles at higher frequencies. Hence, image CTR is actually reduced further (Figs. 6, 7; Table 1).

Table 1. CTR at three areas of the images

CTR* (in linear scale)	MI = 0.05		MI = 0.08	
	1.75 MHz	2.5 MHz	1.75 MHz	2.5 MHz
ROI4/ROI3	7.7 (0.1)	6.8 (0.1)	4.5 (1.0)	5.6 (1.1)
ROI5/ROI6	22.6 (2.5)	8.9 (1.8)	18.6 (3.7)	4.6 (0.8)
ROI1/ROI2	13.6 (1.0)	9.8 (1.2)	11.9 (2.6)	5.5 (0.9)

* All data are presented as mean (SD) of three repeated experiments.

Thirdly, both types of nonlinear propagation cause only a small change in bubble PI signals at the bubble resonance frequency. Nonlinear propagation through overlying tissue makes microbubbles appear slightly brighter in PI images, whereas nonlinear propagation through overlying microbubble clouds makes microbubbles appear slightly darker. The darker appearance of microbubbles is likely a result of the selective attenuation of US by overlying microbubbles at their resonance frequency. For frequencies higher than bubble resonance frequency, both types of nonlinear propagation introduce extra nonlinear components to the scattering from the microbubbles, as it does to tissues, making them appear significantly brighter in the images. This can be seen in Fig. 5, where the relative changes are much higher at 4 MHz and 5.7 MHz. This arises because, away from resonance, the microbubbles behave largely as linear scatterers, and this results in a small denominator in the normalized changes at 4 MHz and 5.7 MHz.

It should be noted that nonlinear propagation is a complex function of spatial location, media properties and parameters of the US scanner. The harmonic components caused by nonlinear propagation build up gradually within the medium (Duck 2002) during propagation until a certain point when they start to decrease because of attenuation. In this study, the simulation results are based on nonlinear propagation through microbubble clouds of ~ 6 cm and through a TMM of 6.5 cm. The CTR results obtained using the clinical scanner correspond to depth of 8.5 cm (ROI 3, 4, 5 and 6) and 3.5 cm (ROI 1, 2), respectively. Furthermore, the effects of nonlinear propagation will also be affected by bubble concentration.

The first part of this study involving transmission measurements and simulation was purposely designed so that confounding factors such as attenuation and noise of the measurement system could be removed or controlled, and the effects of nonlinear propagation through different medium on microbubble nonlinear scattering could be compared with excitation pulses of equivalent power. It should be noted that the simulation results

in this study are based on a single bubble of 6 microns in diameter. Although this corresponds to a typical bubble size present in SonoVue, the effects of bubbles with other sizes are not reflected in the simulation results. The results from the Toshiba prototype scanner, on the other hand, include the effects of bubble population. This partly explains why the simulation results correspond to the practical imaging results only qualitatively.

The second part of this study involving the Toshiba prototype scanner, on the other hand, produces images that contain all the confounding factors including attenuation and spatially variant system noise level. For areas of different acoustic attenuation, bubbles will be excited by pulses of different amplitude. Because microbubble scattering is a nonlinear function of incident pulse amplitude (Fig. 4b.) and tissue scattering is not, CTR will be affected by attenuation. This may contribute to an increased CTR for ROI4/ROI3 because the frequency is changed from 1.75 – 2.5 MHz at an MI of 0.08 (Table 1). As frequency increases, the attenuation for a microbubble cloud is reduced, which can improve the signal from microbubbles under the big microbubble cloud. System noise level can also affect measured CTR when nonlinear propagation is low and the denominator of CTR is influenced by noise. This could explain why the CTR in the lower right area of the images in Fig. 7 was higher than the upper area—the upper area was in the US near-field, where the system signal-to-noise level may be lower than the focal area.

Because nonlinear propagation can significantly affect microbubble nonlinear scattering and CTR, especially at higher frequencies, it is desirable to develop imaging techniques for improved CTR. One method that has had some success in terms of improving CTR, especially at higher frequencies, is that of subharmonic imaging (Goertz *et al.* 2007; Needles *et al.* 2008; Shi *et al.* 1999). This technique explores the difference between the nonlinearity caused by propagation and that resulting from microbubble scattering. Further improvement will probably require the fabrication of bubbles that produce unique nonlinear behavior (Stride *et al.* 2008).

In Fig. 5a, the negative value at 1 MHz for nonlinear propagation through tissue is unexpected. The reason behind this requires further investigation.

In this study, only the PI technique was studied. Although it is likely that the conclusions in this study also apply to other pulsing sequences such as amplitude modulation and pulse inversion amplitude modulation, further studies are needed to confirm that this is the case.

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